

Lesson 17 — Updating & Accumulating Data

Matches LEGO Teacher Portal lesson: "Score Tracker".

Objective (I can...): I can update a variable (increment, accumulate) to track events over time. **CS concept:** Updating variables; accumulation patterns · **Builds on:** Lesson 16 (creating/storing a variable).

AI Connection (2 min): A model keeps running tallies — examples seen, correct predictions, running error. *First-principles question:* "Why does tracking a running total matter more than a single snapshot when something is learning?"

Materials: LEGO CS & AI kit + device per pair, trigger object/button, notebooks. **Model reuse:** Variables model from Lesson 16 — code only.

60-minute flow - Warm-up (8): "Running total" — pairs track two different counts by hand as the coach fires events quickly; connect to "change variable by N." - **Concept + demo (10):** Demo "change score by 1" inside an event so each trigger adds one; show the running total. Then demo accumulating by a sensor reading (add the distance each loop). - **Build/Code (30):** Pairs program a counter (each event adds 1) and a separate accumulator (sum a value over a loop), both displayed. They log the update pattern in the notebook. *Swap at 15 min.* - **Challenge (7):** Add a reset event that sets the counter back to 0 without touching the accumulator. - **Share + pack-up (5):** Pairs demo counting, accumulating, and resetting, and explain the difference between the two. Model stays built.

Pair roles: Driver codes the updates, Navigator fires events and reads the totals. Swap at 15 min. (*Trio rotates code/trigger/read.*)

Differentiation: *Support* — provide the event; students add the "change by 1." *Extension / fast-finisher* — count by a step other than 1, or track two independent counters. *Reteach* — fire one event, watch 0→1, then let them predict the next.

Exit check: Pair's counter and accumulator both update correctly, and the counter resets independently.

At-home (optional): Track something for a day (steps, pages); write it as a variable that gets updated each time. ~10 min.